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#include<iostream>
#include<math.h>
#include<GL/glut.h>
using namespace std;
int xc = 320, yc = 240;
void plot_point(int x, int y)
{
    glBegin(GL_POINTS);
    glVertex2i(xc+x, yc+y);
    glVertex2i(xc+x, yc-y);
    glVertex2i(xc+y, yc+x);
    glVertex2i(xc+y, yc-x);
    glVertex2i(xc-x, yc-y);
    glVertex2i(xc-y, yc-x);
    glVertex2i(xc-x, yc+y);
    glVertex2i(xc-y, yc+x);
    glEnd();
}
void bresenham_circle(int r)
{
    int x=0,y=r;
    float d=3-2*r;
    plot_point(x,y);
    int k;
    while(x < y)
    {
        x = x + 1;
        if(d < 0)
            d=d+4*x+6;
        else
        {
            y = y - 1;
            d=d+4*(x-y)+10;
        }
        plot_point(x,y);
    }
    glFlush();
}
void display()
{
    int radius1=50;
    glClear(GL_COLOR_BUFFER_BIT);
    bresenham_circle(radius1);
}
void Init()
{
    /* Set clear color to white */
    glClearColor(1.0,1.0,1.0,0);
    /* Set fill color to black */

```

```
glColor3f(0.0,0.0,0.0);
gluOrtho2D(0 , 640 , 0 , 480);
}
int main(int argc, char **argv)
{
glutInit(&argc,argv);
glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
glutInitWindowPosition(0,0);
glutInitWindowSize(640,480);
glutCreateWindow("bresenham_circle");
Init();
glutDisplayFunc(display);
glutMainLoop();
}
```